

The Mathematics of HATI

A complete derivation: from a single pixel to a learned hazard probability

Gergő Csaba Morvai

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Abstract

This document derives, from first principles and in full, the mathematics behind HATI’s two terrain-hazard channels. We start with one pixel and a number attached to it, build the discrete differential geometry that turns an elevation grid into ten roughness fields, derive the robust statistics that put those fields on a common footing, and show that the fusion step is exactly a logistic generalized linear model — the same object as a one-neuron classifier. We then develop the full machine-learning apparatus that would *learn* that model (likelihood, cross-entropy, the gradient, the Hessian, regularization, and the backpropagation identity), the exact additive attribution that makes the score interpretable, the projective geometry that places a selenographic coordinate on a pixel, the ray geometry of a cast shadow, the second-moment ellipse algebra of the shadow census, and the sampling theorem that defines the resolution floor both channels are built around. Every block of algebra is followed by a plain-language unpacking. The aim is a single, self-contained derivation you could write across a blackboard and, having understood it, hold the whole pipeline in your head.

Read it once for the shape, once for the algebra, once for the meaning.

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1 The object of study: a discrete field on a lattice

Definition 1.1 (Digital elevation model). Let $\Omega = \{0, \dots, M-1\} \times \{0, \dots, N-1\} \subset \mathbb{Z}^2$ be a finite pixel lattice. A *digital elevation model* (DEM) is a map

$$z : \Omega \longrightarrow \mathbb{R} \cup \{\emptyset\}, \quad (i, j) \mapsto z_{ij},$$

assigning to each pixel a height in metres, or the symbol $\emptyset$ (no data). Each pixel covers a square of side s metres on the ground; s is the *ground sample distance*. The set of valid pixels is $\Omega^* = \{(i, j) \in \Omega : z_{ij} \neq \emptyset\}$.

Definition 1.2 (Affine geotransform). The lattice is tied to projected ground coordinates $(X, Y) \in \mathbb{R}^2$ (metres) by an affine map

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} X_0 \\ Y_0 \end{pmatrix} + \begin{pmatrix} s & 0 \\ 0 & -s \end{pmatrix} \begin{pmatrix} j \\ i \end{pmatrix}, \quad (1)$$

where (X_0, Y_0) is the upper-left corner, column j runs east, row i runs south. The matrix is the *geotransform* G ; it is invertible whenever $s \neq 0$, with G^{-1} recovering (j, i) from (X, Y) .

In words

A DEM is just a grid of numbers: at row i , column j , the ground is z_{ij} metres high. Each cell is s metres across (for our lunar map, $s = 4$ m). Equation (1) is the dictionary between “which cell” and “where on the Moon, in metres.” Everything HAPI does to the terrain is an operation on this grid of numbers; everything that places it on the Moon is this affine map and the projection of the next section.

2 Map I: from a selenographic coordinate to a pixel

The touchdown is given as latitude φ and longitude λ on a sphere of radius $R = 1737.4$ km. We need its pixel. The bridge is the *south-polar stereographic projection*.

2.1 Deriving the stereographic projection

Place the sphere with its south pole S tangent to a plane Π , and project every surface point P from the north pole N through P onto Π . Let θ be the angular distance of P from S (the colatitude measured from the south pole). In the plane through N, S, P , the inscribed-angle relation gives the projected radius

$$\rho(\theta) = 2R \tan\left(\frac{\theta}{2}\right). \quad (2)$$

Latitude relates to this colatitude by $\theta = \frac{\pi}{2} + \varphi$ (at S , $\varphi = -\frac{\pi}{2} \Rightarrow \theta = 0$; at the equator $\varphi = 0 \Rightarrow \theta = \frac{\pi}{2}$). Substituting,

$$\rho(\varphi) = 2R \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right), \quad X = \rho \sin \lambda, \quad Y = \rho \cos \lambda. \quad (3)$$

Proposition 2.1 (Conformality and the point scale). The map (3) is conformal (angle-preserving), and its isotropic point scale factor is

$$m(\varphi) = \frac{d\rho}{R d\theta} = \sec^2\left(\frac{\theta}{2}\right) = \sec^2\left(\frac{\pi}{4} + \frac{\varphi}{2}\right). \quad (4)$$

Proof. Differentiate (2): $\frac{d\rho}{d\theta} = 2R \cdot \frac{1}{2} \sec^2(\theta/2) = R \sec^2(\theta/2)$. The meridian arc element on the sphere is $R d\theta$, so the radial scale is $\sec^2(\theta/2)$. The parallel (circumferential) scale is $\rho/(R \sin \theta)$; using $\rho = 2R \tan(\theta/2)$ and $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$ gives the same $\sec^2(\theta/2)$. Equal radial and circumferential scales $\Rightarrow$ conformal, with common factor (4). At the pole $\theta = 0$, $m = 1$: true scale. $\square$

Finally, compose with the inverse geotransform: the pixel of the touchdown is

$$\begin{pmatrix} j^* \\ i^* \end{pmatrix} = \left\lfloor G^{-1} \begin{pmatrix} \rho(\varphi) \sin \lambda \\ \rho(\varphi) \cos \lambda \end{pmatrix} \right\rfloor, \quad (5)$$

$\lfloor \cdot \rfloor$ rounding to the nearest integer lattice point.

In words

To find which pixel a latitude/longitude lands on, imagine standing at the north pole and shining every south-polar point onto a flat sheet under the south pole. A point's distance from the centre, ρ , depends only on its latitude (Eq. 3); the direction is its longitude. Near the pole the map barely distorts anything — that is what $m \rightarrow 1$ in Eq. (4) means — so a metre on our map is a metre on the Moon. Equation (5) then just reads off the row and column.

3 Differential operators on the pixel lattice

All of Channel 1 is built from two differential primitives — the gradient and the Laplacian — realised as *convolutions* of the grid with small kernels.

Definition 3.1 (Discrete convolution). For a kernel $K \in \mathbb{R}^{(2r+1) \times (2r+1)}$ and field $f : \Omega \rightarrow \mathbb{R}$,

$$(K * f)_{ij} = \sum_{a=-r}^r \sum_{b=-r}^r K_{ab} f_{i-a, j-b}. \quad (6)$$

The first derivatives use central differences (here the Sobel kernels, which average over three rows/columns for noise tolerance):

$$\partial_X z \approx \frac{1}{8s} (K_x * z), \quad \partial_Y z \approx \frac{1}{8s} (K_y * z), \quad K_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, \quad K_y = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}. \quad (7)$$

The factor $1/s$ converts “per pixel” to “per metre.” The *slope field* is the magnitude of the gradient, returned as an angle:

$$\sigma_{ij} = \arctan \sqrt{(\partial_X z)_{ij}^2 + (\partial_Y z)_{ij}^2} \in [0, \frac{\pi}{2}). \quad (8)$$

The second derivative (curvature) uses the five-point Laplacian stencil:

$$(\nabla^2 z)_{ij} \approx \frac{1}{s^2} (K_L * z)_{ij}, \quad K_L = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (9)$$

In words

“Slope” is rise-over-run, measured by comparing each pixel to its neighbours (that’s all a Sobel kernel does — a weighted left-minus-right and up-minus-down). Take the steepness in both directions, combine by Pythagoras, and report the angle: Eq. (8). “Curvature” (Eq. 9) compares a pixel to the average of its four neighbours: positive in a bowl, negative on a dome, zero on a flat or evenly-tilted patch. These two numbers, slope and curvature, are the atoms everything else is built from.

4 Windowed and order statistics

Roughness is not a per-pixel quantity; it lives in a *neighbourhood*. Let $\mathcal{W}_{ij}$ be the $w \times w$ window centred on (i, j) , with $n = w^2$ pixels.

Moment statistics (via the box kernel). With the uniform kernel $U = \frac{1}{n} \mathbf{1}_{w \times w}$, the windowed mean and mean-square of a field f are $\mu_f = U * f$ and $\overline{f^2} = U * f^2$, so the windowed RMS and variance are

$$\text{RMS}_w f = \sqrt{U * f^2}, \quad \text{Var}_w f = (U * f^2) - (U * f)^2. \quad (10)$$

Order statistics. Sort the n window values $f^{(1)} \leq f^{(2)} \leq \dots \leq f^{(n)}$. The p -quantile is $Q_p(f)_{ij} = f^{(\lceil pn \rceil)}$, and the interquartile range is

$$\text{IQR}_w f = Q_{0.75}(f) - Q_{0.25}(f). \quad (11)$$

The median is $\text{med}_w f = Q_{0.5}(f)$. Order statistics are nonlinear — not convolutions — which is exactly why they resist outliers.

In words

A single pixel can’t be “rough.” So we slide a little square window over the map and, at each position, summarise the values inside it. Two kinds of summary: moments (mean, RMS, variance — which we get cheaply as box-blurs, Eq. 10) and order statistics (median, quartiles, IQR — which need sorting, Eq. 11). The order statistics are the robust ones: a single freak value can’t drag a median the way it drags a mean.

5 The ten Tier-1 channels

We now define every channel as an operator $C_k[z] : \Omega \rightarrow \mathbb{R}_{\geq 0}$ on the DEM. Write $\sigma = \sigma[z]$ for the slope field (8) and $\ell = \nabla^2 z$ for the Laplacian (9).

$$\text{RMS slope:} \quad C_1 = \sqrt{U * \sigma^2}, \quad (12)$$

$$\text{Slope IQR:} \quad C_2 = \text{IQR}_w \sigma, \quad (13)$$

$$\text{Curvature IQR:} \quad C_3 = \text{IQR}_w \ell, \quad (14)$$

$$\text{Planar deviation:} \quad C_4 = \sqrt{U * (z - U * z)^2}, \quad (15)$$

$$|\text{TPI}|: \quad C_5 = |z - U * z|, \quad (16)$$

$$\text{TRI:} \quad C_6 = \frac{1}{8} \sum_{(a,b) \in \mathcal{N}_8} |z - S_{ab}z|, \quad (17)$$

$$\text{MDS at scale } L : \quad C_{7...10} = \text{med}_w \left| \sigma[G_{\sigma_L} * z] - \sigma[G_{\sigma_{2L}} * z] \right|, \quad (18)$$

where $S_{ab}z$ is the shift of z by (a, b) , $\mathcal{N}_8$ the eight neighbours, and G_σ the Gaussian kernel $G_\sigma(a, b) = \frac{1}{2\pi\sigma^2} e^{-(a^2+b^2)/2\sigma^2}$ with smoothing radius $\sigma_L = L/3$. The four MDS channels use $L \in \{2, 3, 5, 8\}$ px, i.e. $\{8, 12, 20, 32\}$ m.

In words

Each channel is one “flavour of rough,” all built from slope and curvature inside the sliding window: *RMS slope* = how steep on average; *Slope IQR* = how much the slope *varies in direction* (a crater field scatters slopes every way); *Curvature IQR* = spread of bowl-vs-dome (the crater signature); *Planar deviation* = how bumpy relative to a local flat fit; $|TPI|$ = how far a point sits above or below its surroundings (rim or pit); *TRI* = jaggedness to immediate neighbours; *MDS* = the clever one — blur to size L and to size $2L$, subtract the slopes, and what survives is the roughness living *specifically* at size L . Run it at four sizes and you get a little spectrum of bumpiness across 8–32 m, bracketing the crater that mattered.

6 Robust normalization: putting ten channels on one ruler

The ten fields have different units and wildly different scales. Before combining them we standardise each one against *its own distribution on the scene*.

Tail compression. Roughness fields are non-negative and long-tailed, so we first apply $g(x) = \log(1 + x)$, which is monotone (preserves ranking) and squashes the tail.

Robust location and scale. Instead of mean and standard deviation (which the tail corrupts) we use the median and the IQR. The scale estimate is $\hat{s} = \text{IQR} / 1.349$, and the standardised, clipped channel is

$$\zeta_k = \text{clip}_{\pm c} \left(\frac{g(C_k) - \text{med}(g(C_k))}{\text{IQR}(g(C_k)) / 1.349} \right), \quad \text{clip}_{\pm c}(t) = \max(-c, \min(c, t)), \quad c = 4. \quad (19)$$

Proposition 6.1 (Why 1.349). If $C \sim \mathcal{N}(\mu, s^2)$, then $\text{IQR}(C) = 1.349 s$, so $\text{IQR} / 1.349$ is a consistent estimator of s .

Proof. For the standard normal with CDF Φ , the quartiles are $\pm \Phi^{-1}(0.75)$. Numerically $\Phi^{-1}(0.75) = 0.674489 \dots$, so for $\mathcal{N}(\mu, s^2)$,

$$\text{IQR} = s(\Phi^{-1}(0.75) - \Phi^{-1}(0.25)) = s \cdot 2(0.674489) = 1.34898 s.$$

Dividing by 1.349 inverts this, recovering s . □

In words

Each roughness map is centred (subtract its typical value) and rescaled (divide by its typical spread), so a value of “+2” means the same thing — two spreads above normal — in every channel. We use the median and IQR rather than mean and standard deviation so a handful of extreme pixels can’t define “normal.” The mysterious 1.349 is just the width, in IQRs, of a bell curve (Proposition 6.1): dividing by it turns an IQR back into an ordinary standard deviation. The clip at ± 4 stops any one pixel from screaming over the rest.

7 The fusion is a generalized linear model

Collect the ten standardised channels at a pixel into a *feature vector* $\zeta(x) = (\zeta_1(x), \dots, \zeta_{10}(x))^T \in \mathbb{R}^{10}$, with a literature weight vector $\mathbf{w} \in \mathbb{R}^{10}$ and bias b . HAT’s hazard score is

$$H(x) = \sigma \left(\underbrace{\frac{1}{\|\mathbf{w}\|_1} \mathbf{w}^T \zeta(x) + b}_{a(x) \text{ (pre-activation)}} \right), \quad \sigma(a) = \frac{1}{1 + e^{-a}}, \quad b = -1. \quad (20)$$

This is *exactly* a logistic-regression unit / a single sigmoidal neuron: $\|\mathbf{w}\|_1 = \sum_k |w_k|$ normalises so the output midpoint does not drift with the number of channels. The per-pixel hazard is therefore a point on the logistic curve, driven by a weighted vote of the ten standardised roughness fields.

Exact additive attribution. Because the pre-activation is linear, the score decomposes *exactly* into per-channel contributions

$$a(x) = b + \sum_{k=1}^{10} \phi_k(x), \quad \phi_k(x) = \frac{w_k \zeta_k(x)}{\|\mathbf{w}\|_1}, \quad (21)$$

so “why did this pixel score high?” has a closed-form answer: read off the ϕ_k . For a linear model this coincides with the Shapley value $\phi_k^{\text{Sh}} = \frac{w_k}{\|\mathbf{w}\|_1} (\zeta_k - \mathbb{E}[\zeta_k])$ up to the constant baseline — attribution is not an approximation here, it is the model.

In words

Here is the heart of it. Stack the ten normalised roughness numbers for a pixel into a vector. Multiply by a weight vector (how much each one matters), add them, add a bias, and squash the total through an S-curve into a probability between 0 and 1. That S-curve squashing of a weighted sum *is* the simplest possible classifier — one artificial neuron. And because the inside is just a sum (Eq. 21), you can always say exactly how much each channel pushed the score, which is why HATI is interpretable where a deep network is a black box.

8 The machine learning: learning the weights

HATI fixes $\mathbf{w}$ from the literature. The natural generalization is to *learn* it from labelled terrain. This section is the full apparatus.

8.1 The model as a probability

Treat the label $y \in \{0, 1\}$ (safe / hazardous) as Bernoulli with $\Pr(y = 1 \mid x) = H(x) = \sigma(\mathbf{w}^\top \boldsymbol{\zeta} + b)$. Absorb b into $\mathbf{w}$ by appending a constant feature. Over a dataset $\{(\boldsymbol{\zeta}^{(m)}, y^{(m)})\}_{m=1}^D$ the likelihood is

$$\mathcal{L}(\mathbf{w}) = \prod_{m=1}^D H_m^{y^{(m)}} (1 - H_m)^{1-y^{(m)}}, \quad H_m = \sigma(\mathbf{w}^\top \boldsymbol{\zeta}^{(m)}). \quad (22)$$

8.2 The loss and its gradient

Maximising likelihood = minimising the negative log-likelihood, the *binary cross-entropy*:

$$J(\mathbf{w}) = -\frac{1}{D} \sum_{m=1}^D \left[y^{(m)} \log H_m + (1 - y^{(m)}) \log(1 - H_m) \right]. \quad (23)$$

Using the clean identity $\sigma'(a) = \sigma(a)(1 - \sigma(a))$, the gradient collapses beautifully:

$$\begin{aligned} \frac{\partial J}{\partial \mathbf{w}} &= -\frac{1}{D} \sum_m \left[\frac{y^{(m)}}{H_m} - \frac{1 - y^{(m)}}{1 - H_m} \right] \sigma'(a_m) \boldsymbol{\zeta}^{(m)} \\ &= -\frac{1}{D} \sum_m \left[\frac{y^{(m)} - H_m}{H_m(1 - H_m)} \right] H_m(1 - H_m) \boldsymbol{\zeta}^{(m)} = \boxed{\frac{1}{D} \sum_{m=1}^D (H_m - y^{(m)}) \boldsymbol{\zeta}^{(m)}}. \end{aligned} \quad (24)$$

Gradient descent then iterates $\mathbf{w} \leftarrow \mathbf{w} - \eta \partial_{\mathbf{w}} J$ with learning rate η .

8.3 Curvature: the Hessian and Newton's method

Differentiating (24) again,

$$\mathbf{H} = \frac{\partial^2 J}{\partial \mathbf{w} \partial \mathbf{w}^\top} = \frac{1}{D} \sum_m H_m(1 - H_m) \boldsymbol{\zeta}^{(m)} \boldsymbol{\zeta}^{(m)\top} = \frac{1}{D} \Phi^\top S \Phi \succeq 0, \quad (25)$$

with design matrix $\Phi = [\boldsymbol{\zeta}^{(1)}, \dots]^\top$ and $S = \text{diag}(H_m(1 - H_m))$. Positive semi-definiteness proves J is *convex*: no spurious local minima. Newton's method (here, iteratively reweighted least squares) updates $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{H}^{-1} \partial_{\mathbf{w}} J$.

8.4 Regularization

A zero-mean Gaussian prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 I)$ turns maximum likelihood into maximum-a-posteriori, adding the ridge penalty

$$J_\lambda(\mathbf{w}) = J(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2, \quad \lambda = \frac{1}{D\tau^2}, \quad (26)$$

which keeps weights small (less overfitting) and makes $\mathbf{H} + \lambda I \succ 0$ strictly — a unique optimum.

8.5 Backpropagation to a deeper model

If the ten channels feed a hidden layer rather than a single neuron, write $\mathbf{h} = \sigma(W^{[1]}\boldsymbol{\zeta} + \mathbf{b}^{[1]})$, $H = \sigma(\mathbf{w}^{[2]\top} \mathbf{h} + b^{[2]})$. The chain rule (backprop) propagates the error $\delta^{[2]} = H - y$ backward,

$$\delta^{[1]} = (\mathbf{w}^{[2]} \delta^{[2]}) \odot \mathbf{h} \odot (1 - \mathbf{h}), \quad \frac{\partial J}{\partial W^{[1]}} = \delta^{[1]} \boldsymbol{\zeta}^\top, \quad (27)$$

$\odot$ the elementwise product — the same $\sigma' = \sigma(1 - \sigma)$ doing the work at every layer. (For k mutually-exclusive hazard classes one replaces σ by the softmax $\text{softmax}(a)_c = e^{a_c} / \sum_{c'} e^{a_{c'}}$ and the loss by categorical cross-entropy; the gradient keeps the same $H - y$ form.)

In words

This is the part everyone calls “training.” Say which sites are safe ($y=0$) and which are hazardous ($y=1$). The model guesses a probability H ; the cross-entropy (Eq. 23) measures how wrong the guesses are. The miracle is Eq. (24): the nudge each weight needs is just (*prediction minus truth*) *times the feature* — push the weight up if you under-predicted hazard on a rough feature, down if you over-predicted. Repeat. The Hessian (Eq. 25) being positive shows the problem is bowl-shaped — one bottom, no traps. Regularization keeps the weights modest; backprop is the same rule applied layer by layer if you stack neurons. HAT1 currently *skips* this step and sets the weights from physics — which is what makes it interpretable and needs no training labels — but this is the door to learning them.

9 From a score field to a decision

The hazard field $H : \Omega^* \rightarrow (0, 1)$ is *relative*. To rank a point we use the empirical cumulative distribution over valid pixels,

$$\hat{F}_H(t) = \frac{1}{|\Omega^*|} \sum_{x \in \Omega^*} \mathbf{1}\{H(x) \leq t\}, \quad (28)$$

and report the touchdown's percentile $\hat{F}_H(H(x^*))$. As a unit-free effect size we also give the robust z against the median absolute deviation $\text{MAD} = \text{med}_x |H(x) - \text{med } H|$,

$$z^* = \frac{H(x^*) - \text{med}_x H}{1.4826 \text{MAD}}, \quad (29)$$

($1.4826 = 1/\Phi^{-1}(0.75)$ rescales MAD to a Gaussian σ). The decision rule is $\text{flag}(x) = \mathbf{1} \left\{ \widehat{F}_H(H(x)) \geq \tau \right\}$ for a chosen percentile threshold τ .

In words

A score of 0.73 means nothing on its own — is that bad? So we ask: *what fraction of the scene is less hazardous than this point?* That percentile is the verdict. The robust z says the same thing in “how many spreads above typical.” For the Athena touchdown both came out extreme (96th percentile, $z^* = 3.2$), which is what “flagged” means.

10 Channel 2: the geometry of a cast shadow

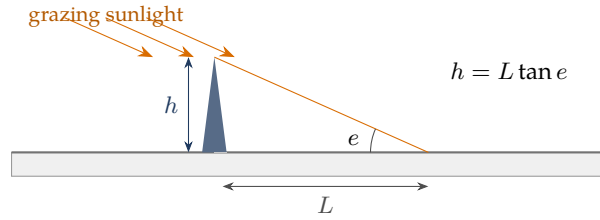
The DEM stops at the pixel scale. Channel 2 reaches below it using shadows in a finer photograph.

10.1 Height from shadow length

Let the Sun sit at elevation angle e above the local horizon. An obstacle of height h blocks the grazing rays; the shadow it casts on flat ground has length L where, by the right triangle of ray, ground, and obstacle,

$$\tan e = \frac{h}{L} \implies \boxed{h = L \tan e.} \quad (30)$$

The shadow’s *width across* the Sun direction equals the obstacle’s footprint and is independent of e (the Sun’s elevation stretches the shadow only along its own azimuth). So one shadow yields two numbers: an elevation-free footprint, and a height that needs e .



10.2 Detecting shadows under a varying illumination gradient

A global brightness threshold fails at the pole (slopes facing away from the Sun are dark without being shadows). Define instead a *local-relative* shadow mask on the ortho-image I :

$$\mathcal{S}_{ij} = \mathbf{1} \{ 0 < I_{ij} < \alpha \cdot (\text{med}_{45} I)_{ij} \}, \quad \alpha = \frac{1}{2}, \quad (31)$$

med_{45} a 40 m median filter giving the local sunlit background. Morphological opening then closing cleans speckle and pinholes.

10.3 Measuring each shadow: the second-moment ellipse

Label the connected components of $\mathcal{S}$. For a region $\mathcal{R}$ with pixel coordinates $\{\mathbf{p}_k\}$, centroid $\bar{\mathbf{p}}$, form the *second-moment (inertia) matrix*

$$\mathbf{M} = \frac{1}{|\mathcal{R}|} \sum_{\mathbf{p} \in \mathcal{R}} (\mathbf{p} - \bar{\mathbf{p}})(\mathbf{p} - \bar{\mathbf{p}})^\top = \begin{pmatrix} \mu_{20} & \mu_{11} \\ \mu_{11} & \mu_{02} \end{pmatrix}. \quad (32)$$

Its eigenvalues $\lambda_1 \geq \lambda_2$ are

$$\lambda_{1,2} = \frac{\mu_{20} + \mu_{02}}{2} \pm \sqrt{\left(\frac{\mu_{20} - \mu_{02}}{2} \right)^2 + \mu_{11}^2}, \quad (33)$$

and the equivalent ellipse has axis lengths and orientation

$$\underbrace{\Lambda_{\max} = 4\sqrt{\lambda_1}}_{\text{along-Sun: } L}, \quad \underbrace{\Lambda_{\min} = 4\sqrt{\lambda_2}}_{\text{cross-Sun: footprint}}, \quad \beta = \frac{1}{2} \arctan \frac{2\mu_{11}}{\mu_{20} - \mu_{02}}. \quad (34)$$

A footprint $\Lambda_{\min} s_I$ (with ortho scale $s_I = 0.9$ m) below the DEM floor (Sec. 11) is an obstacle the DEM cannot see.

10.4 Internal validity: circular statistics of orientation

Shadow orientations $\{\beta_r\}$ are *axial* (mod π). Their concentration is measured by the doubled-angle resultant

$$\bar{\beta} = \frac{1}{2} \arg \left(\sum_r e^{i2\beta_r} \right), \quad \mathcal{R} = \frac{1}{N} \left| \sum_r e^{i2\beta_r} \right| \in [0, 1]. \quad (35)$$

$\mathcal{R} \rightarrow 1$ (a single peak) certifies the dark patches are illumination-driven shadows, not random albedo — which has no preferred direction.

10.5 Multi-angle height: an over-determined solve

Two or more frames at solar elevations $\{e_t\}$ give shadow lengths $\{L_t\}$ for the same obstacle. The height is the least-squares solution of $L_t \tan e_t = h$:

$$\hat{h} = \arg \min_h \sum_t (L_t \tan e_t - h)^2 = \frac{1}{T} \sum_{t=1}^T L_t \tan e_t, \quad (36)$$

a stereo-free relief estimate (no image matching, hence independent of the DEM's correlation errors).

In words

At the pole the Sun barely clears the horizon, so a knee-high rock throws a long shadow — a free ruler for things too small for the DEM. The shadow's length gives the rock's height (Eq. 30); its width gives the footprint. We find shadows by calling a pixel “dark relative to its neighbourhood” (not just “dark,” which polar slopes already are), then fit an ellipse to each dark blob using the same covariance-eigenvalue trick a statistician uses to draw a data cloud's axes. If all the shadows point the same way, they're real shadows, not stains. And with two photos at different Sun angles you can solve the height outright — without ever touching the stereo DEM, which is why it's the clean cross-check.

11 The resolution floor, and fusing the two channels

Proposition 11.1 (Nyquist floor). A DEM sampled at spacing s can represent surface undulations only up to spatial frequency $f_{\text{Ny}} = \frac{1}{2s}$; the shortest resolvable wavelength is $\lambda_{\min} = 2s$.

Proof. Sampling $f(\mathbf{X})$ at spacing s replicates its spectrum $\hat{f}$ at every multiple of $1/s$. Components above $1/2s$ fold back (alias) onto lower frequencies and are unrecoverable; by the sampling theorem exact recovery requires the signal be band-limited below $1/2s$. Hence $\lambda_{\min} = 2s$. $\square$

For $s = 4$ m this is 8 m: the DEM is blind below it. The 0.9 m ortho's shadows reach far beneath, which is the whole reason for two channels. Because the ortho is co-registered to the DEM by the same projection (Sec. 2), the shadow-derived sub-resolution relief h can be rasterised onto the DEM grid and fused. The two channels answer different questions, so they are combined not by naive summation but as a resolved-scale rank and an absolute sub-resolution floor:

$$\text{Hazard}(x) = \max \left(\hat{F}_H(H(x)), \Psi(\rho_{\text{sub}}(x)) \right), \quad (37)$$

with ρ_{sub} the local density of sub-floor obstacles and Ψ a calibrated monotone map.

In words

Cut a photograph into 4 m squares and anything smaller than about two squares (8 m) blurs away — that’s the sampling theorem, the same reason a low audio sample rate loses the high notes. So the DEM is structurally blind below 8 m. The shadow channel, on a four-times-finer photo, sees right into that blind band. Stack the two: the DEM ranks the resolved roughness, the shadows set an absolute floor of “stuff too small to map,” and the hazard is whichever alarm is louder.

12 Synthesis: the whole pipeline in one line

Composing every operator above, HATI’s resolved-scale hazard field is the single map

$$H(x) = \sigma \left(b + \frac{1}{\|\mathbf{w}\|_1} \sum_{k=1}^{10} w_k \text{clip}_{\pm c} \left[\frac{g(C_k[z])(x) - \text{med } g(C_k[z])}{\text{IQR } g(C_k[z])/1.349} \right] \right) \quad (38)$$

where z is the DEM, C_k the ten convolution/order-statistic operators of Sec. 5, $g = \log(1 + \cdot)$, σ the logistic, and $(\mathbf{w}, b)$ either literature-fixed (HATI today) or learned by the logistic regression of Sec. 8. Alongside it, Channel 2 produces the shadow census $\{(\Lambda_{\min} s_I, L_t \tan e_t)\}$ of Sec. 10. Together they map an elevation grid and a photograph to a calibrated, *interpretable*, training-free statement about where a lander should not touch down.

In words

Equation (38) is the entire resolved-scale pipeline as one symbol. Read it inside-out: take the DEM z ; hit it with ten roughness operators C_k ; log-squash and standardise each (g , median, IQR, clip); take a weighted sum; add a bias; squash through the S-curve. Out comes a hazard probability for every pixel. Every piece of that line was derived above, and every piece has a one-sentence meaning. If you can read this equation and narrate it, you understand HATI.